

Factory Monitoring & Predictive Maintenance System

Final Project Proposal

1. Project Description

The **Factory Monitoring & Predictive Maintenance System** is an intelligent platform designed to monitor factory machines, analyze operational data, and predict potential failures before they occur. The goal of the system is to improve machine reliability, reduce unexpected downtime, and help factories make better operational decisions.

The system will analyze multiple datasets related to machine operations, including **machine information, failure records, maintenance history, and machine error logs**. By analyzing these datasets, the platform will identify patterns that indicate potential machine problems and provide early warnings before failures occur.

The platform will include a monitoring interface that allows users to track machine performance, efficiency, and operational status. Through interactive dashboards, users will be able to view important metrics such as machine utilization, error frequency, maintenance history, and failure risk levels.

In addition, the system will include an **image classification model** that analyzes machine images to determine whether visible defects or damages are present. This module will use CNN techniques to classify machine conditions.

The project also aims to support decision-making in factory operations. For example, if production needs to increase, the system can analyze machine data to estimate which machines are at higher risk of failure and whether increasing production could negatively impact machine performance or maintenance costs.

A future extension of the platform will include an **AI-powered chatbot assistant** that can answer questions about machine conditions, explain possible causes of failures, and provide recommendations for maintenance or operational adjustments.

Overall, the system will help factories move toward smarter manufacturing by combining **data analysis, machine learning, and visualization tools** to improve productivity and reduce operational risks.

2. Group Members

- Shehabeldeen Yousef Hassan
 - Ahmad Achraf Abdelbary
 - Ahmad Abdel Latif Amer
 - Shahd Ahmed Khaled
 - Ziad Samy Said
 - Saad Salah
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3. Team Leader

The team leader will be responsible for organizing the workflow, ensuring that project milestones are achieved on time, and coordinating communication between all team members.

4. Objectives

The main objectives of the project are:

- Monitor factory machines and track their operational performance.
 - Analyze historical machine data to identify patterns and trends.
 - Predict possible machine failures using machine learning models.
 - Identify the most influential factors that lead to machine breakdowns.
 - Detect machine defects using image classification techniques.
 - Provide dashboards for monitoring machine status and performance.
 - Support decision-making regarding machine usage and production planning.
 - Reduce unexpected machine downtime and maintenance costs.
 - Improve operational efficiency in factory environments.
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5. Tools & Technologies

Programming & Data Processing

- Python
- Pandas
- NumPy

Machine Learning

- Scikit-learn
- TensorFlow or PyTorch

Data Visualization

- Power BI
- Matplotlib
- Seaborn
- Dashboard

Application Development

- Streamlit

Backend Development

- FastAPI

Database

- SQL or SQLite

These technologies will be used to build the complete system, from data processing and model development to visualization and platform deployment.

Milestone	Description	Deadline
Project Planning	Define project scope, objectives, and overall system architecture	7 April 2026
Data Preparation	Collect, clean, and preprocess machine datasets for analysis	27 April 2026
Exploratory Data Analysis	Analyze data patterns and understand machine behavior	4 May 2026
Feature Engineering	Create meaningful features for machine learning models	11 May 2026
Model Development	Train and evaluate machine failure prediction models	18 May 2026
Image Classification Model	Develop an image classification model for defect detection	25 May 2026
Dashboard Development	Build dashboards to monitor machine performance and system insights	20 June 2026
System Integration	Integrate data processing, machine learning models, and user interface	27 June 2026
Testing & Optimization	Test the system and optimize overall performance	5 July 2026
Final Presentation	Prepare the final demonstration and complete the project report	20 July 2026

7. KPIs (Key Performance Indicators)

1. Data Quality

Metric	Target
Percentage of missing values handled	95%
Data accuracy after preprocessing	90%
Dataset diversity (representation of different categories)	85%

2. Model Performance

Metric	Target
Model Accuracy / F1 Score	87%
Model prediction speed (Latency)	< 100 milliseconds
Error rate (False Positive / False Negative)	< 10%

3. Deployment & Scalability

Metric	Target
API uptime	99%
Response time per request	< 200 milliseconds
Real-time image processing speed	10 frames per second

4. Business Impact & Practical Use

Metric	Target
Reduction in manual monitoring effort	60%
Expected cost savings	25–30%
User satisfaction	90%